A DIGITAL LOOP FILTER FOR A PHASE LOCKED LOOP

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ABSTRACT

Modern digital telecommunication and audio systems include a Digital Phase Locked Loop (D-PLL) in a form of a device or an algorithm. Wireless infrastructure, broadband wire-line networks and high end audio systems require very high performance PLLs. There are several key points required for high performance D-PLL circuits design such as power efficiency, loop bandwidth flexibility and accurate frequency translation. This paper describes a novel D-PLL architecture and presents an analysis of the digital loop filter.

1. INTRODUCTION

In digital wireless communication systems (GSM, UMTS etc.), PLLs are used to provide the local oscillator for up-conversion during transmission and down-conversion during reception and clock generation. Frequency synthesizers have to operate over a frequency tuning range of several hundred of MHz with frequency steps of few hertz. Loop bandwidths of few hertz may need to be used to minimize spurs and settling time in channel switching has to be in the order of μsec serving needs for fast channel switching. Transceivers' local oscillator modules are typically built with a PLL based frequency synthesizer integrated circuit [1]. Analog phase locked loops are generally built with an analog phase detector, low pass filter, a VCO placed and a divider in a negative feedback configuration (Figure 1a). Various types and orders of analog VCO based PLLs have been analyzed. Second-order analog PLLs have been analyzed by Hein and Scott [2] and Gardner [3]. Several other references [4], [5], [6] provide an analysis and design procedure for third-order charge-pump PLLs.

In a PLL a phase detector compares two input signals and produces an error signal which is proportional to their phase difference. The error signal is then low-pass filtered and used to drive a VCO which creates an output phase. The input is called the reference signal (f_{REF} in figure 1a). The output is fed through a divider back to the input of the system, producing a negative feedback loop. If the output phase drifts, the error signal will increase, driving the VCO signal phase in the opposite direction so as to reduce the error. Thus the output signal phase is locked to the phase at the other input signal.

The last decade several digital and digital phase-locked loops (PLLs) for various applications were reported [7]–[10]. They demonstrate the ability of a digital implementation to achieve the performance of analog PLLs and even outperform them. There are several other advantages of a digital implementation of PLLs. These include eliminating the noise-susceptible analog control for a voltage-controlled oscillator (VCO) and the inherent noise immunity of digital circuits.

A digital phase locked loop uses a digital phase detector, a digitally controlled oscillator (DCO) or a direct digital synthesizer (DDS), a divider in the feedback path and a divider in the reference path, in order to make the PLL's output signal frequency a rational multiple of the reference frequency (Figure 1b) [11]. A non-integer multiple of the reference frequency can also be created by replacing the simple divide-by-N counter in the feedback path with a programmable pulse swallowing counter. This technique is usually referred to as a fractional-N synthesizer or fractional-N PLL.

In an analog PLL, the phase detector generates charge-pump current pulses, whose duration is proportional to the phase difference between the reference and feedback signal. In a digital PLL, on the other hand, the phase detector's output is a digital number proportional to the time difference between the edges of the incoming reference signal and the feedback signal. These digital words are sent to the digital loop filter, which filters and integrates the phase detector output. Because the loop filter parameters are numerical coefficients, they can be easily modified, and unlike an analog PLL, they have more flexible limit to how large they can be. In addition, the digital phase detector doesn't suffer from thermal noise, aging or drift, and charge pump mismatch or leakage. The last advantage is particularly important as charge-pump leakage and driver

up/down current mismatch are key contributors to spurs that appear in the output spectrum at the phase detector frequency. A digital PLL avoids this by eliminating the charge pump. The Phase Frequency Detector (PFD) detects the phase and frequency mismatch of the reference signal and divided DCO signal. The PLL is locked when the PFD detects that the phase and frequency of the two signals inputs match. The output of the PFD drives the time to digital converter. The PFD produces up and down enable signals that are interfaced to the time to digital converter. The time to digital converter takes these inputs and increases or decreases the control word which is fed to the thermometric decoder. This decoder is essential for controlling the DCO. The DCO signal is divided by a specific factor and sent back to the PFD for phase and frequency comparison. However, it is reported [12], [13] [14] that that about 60% power consumption of a DCO based D-PLL comes from the DCO itself.

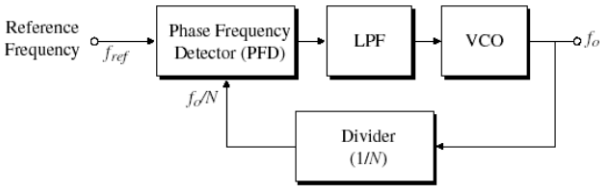


Figure 1: Phase Locked Loop

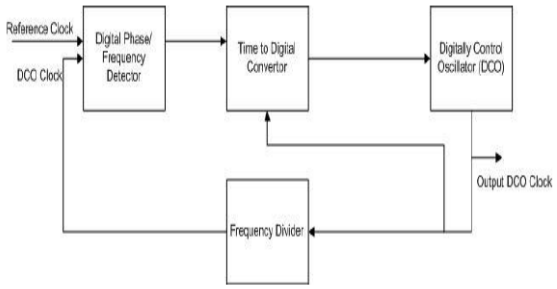


Figure 2. DCO based D-PLL.

2. NOVEL DIGITAL PHASE LOCKED LOOP

This paper presents a D-PLL architecture based on a Direct Digital synthesizer (DDS) instead of a DCO. The proposed architecture enhances D-PLL performance due to DCO's tuning range and power consumption limitations. The proposed D-PLL consists of a phase frequency detector, a loop filter, a cascaded comb integrator (CCI) filter, a direct digital synthesizer (DDS), a feedback divider, a reconstruction filter and control logic. These functional blocks are shown in a block diagram in Figure 3. The novelty of this architecture is the implementation of the loop filter that is a second order digital filter cascaded by a CCI filter. The phase frequency detector and the loop filter are implemented using digital techniques, which offer advantages over conventional analog designs. The operation of the loop filter is explained in the next section of this paper. The CCI rate converter filter provides the necessary sample rate conversion between the sample rate of the loop filter and the sample rate of the DDS. The rate converter suppresses the spectral images that are a consequence of the rate conversion process. The feedback divider is a programmable integer divider that reduces the DDS output frequency to that of f_{REF}. This allows the DDS to run at a rate that is an integer multiple of f_{REF}. The control logic connects the loop elements together functionally. The control loop block diagram is shown in Figure 3.

The timing control logic is triggered by phase detector block. The sample rate of the loop filter is greater than the f_{REF} rate. This is established by appropriately programming the "P" divider. Hence, the loop filter over -samples the output of the phase detector. The loop filter operates at a fixed rate of f_S/P. The Phase Frequency Detector, on the other hand, operates at a different rate, that is a sub-multiple of f_S as well.

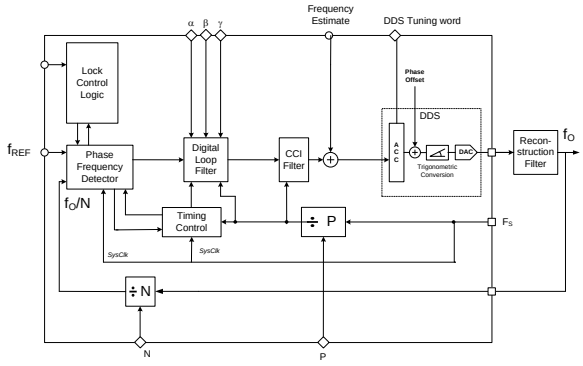


Figure 3: DDS based Digital Phase Locked Loop.

3. DIGITAL LOOP FILTER

The functional diagram of the Digital Loop Filter is shown in Figure 4. The digital implementation of the loop filter is intended to emulate the response of the passive analog second order filter shown in figure 4. The relationship between the analog components and the digital filter coefficients is detailed in section VI of this paper.

The blocks labeled "Σ" in the diagram are digital integrators. The z-domain response of an integrator is given by:

$$H_{INT}(z) = \frac{1}{z-1} \qquad (1)$$

Given the z-domain response of each integrator, it can be shown that the z-domain response of the loop filter is:

$$H_{IIR}(z) = \alpha\left(\frac{z+(\beta-\gamma-1)}{z^2+(-\gamma-2)z+(\gamma+1)}\right) \qquad (2)$$

where $z = e^{j2\pi P\frac{f}{f_S}}$

The sample rate associated with z is f_S/P, where P is the rate conversion factor of the CCI filter.

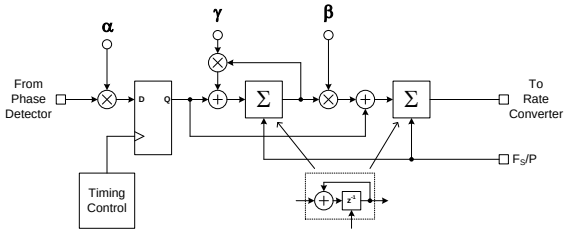


Figure 4: Digital Loop Filter Diagram

The $H_{IIR}(z)$ equation indicates that the denominator is of second order, so there are two poles.

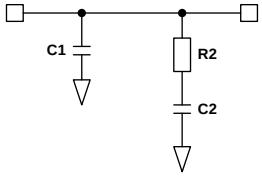


Figure 5: Second order Loop Filter.

The poles are determined by equating the denominator with zero and solving for z. Hence, the pole locations are $p_1 = 1$ and $p_2 = 1 + \gamma$. The $H_{IIR}(z)$ equation also indicates that the numerator has a single response zero. The zero is determined by equating the numerator with zero and solving for z. Hence, the zero location is: $z_1 = 1 + \gamma - \beta$. Finally, the $H_{IIR}(z)$ equation indicates that there is a scale factor established by the coefficient α.

A floating point implementation of the loop filter is not practical as most digital devices use fixed point implementation. Some trade-offs must be made between the granularity of the loop filter coefficients (α, β, γ) and the ideal response characteristics of the loop filter. One of these trade-offs is to constrain both β and γ to values less than -0.125 ($-1/2^3$). This constraint puts a limitation on the maximum relative loop bandwidth (f_C/f_S) that can be accommodated and sets the ceiling for phase margin to something less than 90°. Both the β and γ coefficients are implemented as a two-stage multiplication. The first stage is a right-shift that provides a coarse multiply of 2^{-W}. ($0 \le w \le 3$). The second stage is a 12-bit multiplier that scales the shifted value. The α coefficient is partitioned into two separate stages. The first stage is a "left-shift" that implements a multiplication factor of 2^Y ($0 \le Y \le 15$). The left-shift stage is located at the input to the loop filter. The second stage of the α coefficient is located at the output of the loop filter and is a 12-bit multiplier that scales the shifted value.

4. LOOP FILTER CLIPPING

The output of the loop filter represents an offset to the nominal frequency control word supplied to the DDS. Therefore, its output is centered at zero and extends to its maximum positive and negative extremes. If the loop filter output were to over- or underflow, it would appear as a very large step function at the frequency control input of the DDS. The design must prevent the loop filter output from either over-flowing or under-flowing. Therefore, the loop filter output must be clipped at its positive and negative extremes. The clipping function operates on the final integrator stage of the loop filter. The clipping logic monitors the present output state of the integrator and the next input sample. If their sum exceeds the positive or negative range of the integrator then the integrator is set to the appropriate clip level (i.e., maximum positive or negative).

5. CASCADE COMB-INTEGRATOR (CCI) FILTER

The purpose of a CCI filter is to take the loop filter output samples, which occur at a rate of f_S/P, and interpolate intermediate samples to produce output samples at a rate of f_S. (f_S is the DDS frequency operation). Additionally, the CCI filter provides image suppression to minimize the consequences of the Nyquist images that result from sample rate up-conversion. Rejection of Nyquist images is important for reducing wideband noise in the output spectrum in order to help meet the overall phase noise and jitter performance. A block diagram of the CCI filter is shown in Figure 5. It is a second order filter, which means that it has two comb and two integrator stages. The CCI filter is to be designed with P restricted to power of 2 values. This eliminates the insertion loss associated with non power of 2 values of P.

The CCI filter of Figure 6 is somewhat different than a conventional CCI. One comb stage and one integrator stage are not shown because a single comb stage followed by an integrator stage with zero-stuffing results in an oversampled, but exactly the same of the input data stream. The CCI filter has a 48-bit input bus and a 64-bit output bus. The increase in required bus width is a consequence of the fact that the integrator operates at a sample rate that is greater than the comb stage by a factor of P. Since the rate ratio P can be as large as 2^{16} in this PLL, the bus width of the final integrator must be 64-bits wide.

Since the output of the CCI filter is intended to serve as a 48-bit tuning word adjustment to the DDS, then only 48 significant bits need be extracted from the CCI output. The most important aspect of the CCI filter is its frequency response. Its purpose is not only to provide the necessary rate conversion, but to suppress the accompanying spectral images, as well.

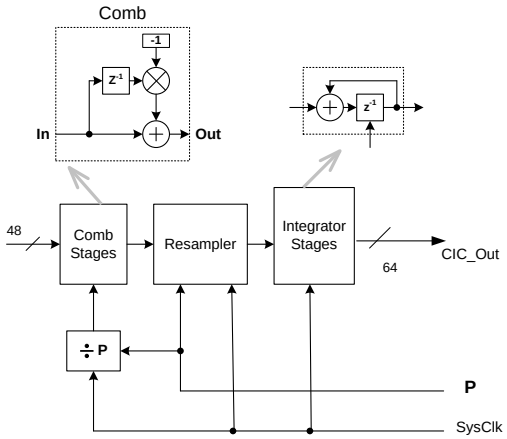


Figure 6: Cascaded Comb-Integrator (CCI) Filter

The response characteristic of the CCI is given by:

$$H_{CCI}(z) = \left[\frac{1 - z^{-P}}{P(1 - z^{-1})} \right]^2 \qquad (3)$$

Where P is the sample rate conversion factor. The sample rate associated with z in the H_{CCI} formula is f_S. Hence, z can be evaluated in terms of frequency as:

$$z = e^{j2\pi \frac{f}{f_S}} = e^{j\omega} \qquad (4)$$

Thus, the transfer function in the frequency domain is given by:

$$H_{CCI}(j\omega) = \left[\frac{1 - e^{-j\omega P}}{P(1 - e^{-j\omega})} \right]^2 \qquad (5)$$

The magnitude response can then be shown to be:

$$\left| H_{CCI}(j\omega) \right| = \frac{1}{P^2} \left(\frac{1 - \cos(\omega P)}{1 - \cos(\omega)} \right) \qquad (6)$$

The magnitude response characteristic associated with $H_{CCI}(j\omega)$ is a series of lobes and null points. The response is maximum at DC and f_S. The null points occur at multiples of $\omega = 2\pi f_S/P$. This is shown in Figure 6 for the case, P=8. The frequency scale is relative to the input sample rate of the CCI filter (f_S/P). Keep in mind that f_S is 8 times the input rate.

The first lobe monotonically decreases until it reaches the first null point at frequency at f=1, which corresponds to f_S/P. Subsequent lobes exhibit successively lower maxima until the Nyquist frequency is reached at f=(1/2)P=4, which corresponds to $(1/2)f_S$. The spectrum is mirrored about the Nyquist frequency so that a maximum again appears at f=P=8, which corresponds to f_S.

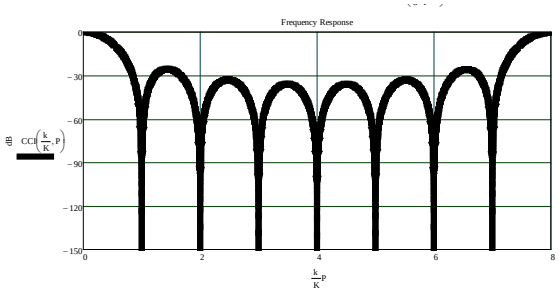


Figure 7: CCI Filter frequency response.

In a previous section it was mentioned that the CCI response is maximum at DC. This fact is not readily apparent from the magnitude response function, $|H_{CCI}(j\omega)|$. Note that at DC, $\omega=0$, so we are interested to evaluate $|H_{CCI}(j0)|$. An application of l'Hopital's Rule yields the result:

$$\left| H_{CCI}(j0) \right| = 1 \qquad (7)$$

This information is useful for determining the cutoff frequency (ω_c) of the CCI filter. The cutoff frequency of a filter is defined at the -3dB point. Since the CCI response is maximum at DC ($\omega=0$), the -3dB point occurs when the square of the magnitude response (i.e., the power response) is half of the maximum value, which can be expressed as:

$$\left| H_{CCI}(j\omega_c) \right|^2 = \frac{1}{2} \left| H_{CCI}(j0) \right|^2 \qquad (8)$$

Since $|H_{CCI}(j0)|=1$, we have:

$$\left[\frac{1}{P^2} \left(\frac{1 - \cos(\omega_c P)}{1 - \cos(\omega_c)} \right) \right]^2 = \frac{1}{2} \qquad (9)$$

Solving for ω_c in the above equation yields:

$$f_c = \frac{f_S}{2\pi} \cos^{-1} \left[1 - \frac{\sqrt{2}}{P^2} + \frac{\sqrt{2}}{P^2} \cos(2\pi P \frac{f_c}{f_S}) \right] \qquad (10)$$

where $\omega_C = 2\pi\, f_C / f_S$.

6. LOOP FILTER ANALYSIS

The loop filter is functionally composed of two independent filters. The first is a second order Infinite Impulse Response (IIR) filter that emulates the response of a typical second order analog low-pass loop filter. This is followed by a CCI filter that provides the sample rate conversion between the loop filter and the DDS. The composite response of the loop filter can be expressed as:

$$H_{LF} = H_{IIR} G_{CCI} H_{CCI} \qquad (11)$$

With regard to the CCI filter its response is the product of two terms – an insertion loss term, G_{CCI}, and a frequency dependent term, H_{CCI}. The H_{CCI} exhibits a lobe that extends from DC to the first null point (which occurs at a frequency that is equal to the CCI input sample rate). The PLL loop bandwidth is chosen to be at least an order of magnitude smaller than the input sample rate of the CCI filter. This places the PLL loop bandwidth in a very narrow region at the beginning of the first lobe of the CCI response. In this region the magnitude response can be considered flat and exhibits approximately unity gain. Therefore, we can use the approximation, $H_{CCI}=1$, which allows us to ignore its contribution to the overall loop response. Hence, H_{LF} simplifies to:

$$H_{LF} = H_{IIR} G_{CCI} \qquad (12)$$

where $G_{CCI} = 2^{\log_2 P - ceil(\log_2 P)}$). When $P=2^x$ and "x" is an integer, then $G_{CCI}=1$.

The open loop gain response is given by the equation:

$$H_{OL} = \left[K_{PD} H_{IIR} G_{CCI} \left(\frac{G_{DDS}}{z-1} \right) \right] \qquad (13)$$

Where G_{DDS} is the DDS gain and K_{PD} is the phase frequency detector gain.

The goal is to design the loop filter response (define coefficients α, β, and γ in equation 2) based on the open loop magnitude and phase response $|H_{OL}|$ and ϕ_{OL}, respectively. The magnitude response provides us with the gain-bandwidth as a metric, and the phase response provides us with the phase margin as a metric. The loop bandwidth, is the radian frequency, ω_1 ($\omega_1=2\pi f_1$), at which $|H_{OL}|=1$. The phase margin, ϕ_M, is the amount by which ϕ_{OL} deviates from -180° at the loop bandwidth frequency f_1.

As mentioned in the section III, the loop filter response is designed to emulate a second order low-pass filter made up of C_1, C_2, and R_2 as shown in Figure 4. It can be shown that the analog components of an analog PLL (figure 1a) are given by [1]:

$$C_1 = \frac{f_R K_{PD} f_s K_{VCO}}{\omega_1^2 N} \sqrt{\frac{1-\sin(\varphi_M)}{1+\sin(\varphi_M)}} \qquad (14)$$

$$C_2 = \frac{2 f_R K_{PD} f_s K_{VCO} \tan(\varphi_M)}{\omega_1^2 N} \qquad (15)$$

$$R_2 = \frac{\omega_1 N}{2 f_R K_{PD} f_s K_{VCO}} \left(1 + \frac{1}{\sin(\varphi_M)} \right) \qquad (16)$$

It can also be shown that C_1, C_2 and R_2 are related to α, β and γ by:

$$\alpha = \left(\frac{P f_R}{f_S} \right) \left(\frac{1}{C_1} \right) = \frac{P \omega_c^2 N}{K_{PD} K_{VCO}} \sqrt{\frac{1+\sin(\varphi_M)}{1-\sin(\varphi_M)}} \qquad (17)$$

$$\beta = \left(-\frac{P}{f_S} \right) \left(\frac{1}{C_1 R_2} \right) = -2P\omega_c \tan(\varphi_M) \qquad (18)$$

$$\gamma = \left(-\frac{P}{f_S} \right) \left[\frac{1}{R_2} \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \right] = -P\omega_c \left(\frac{1+\sin(\varphi_M)}{\cos(\varphi_M)} \right) \qquad (19)$$

Where $\omega_c = \frac{\omega_1}{f_S}$

This means that for a given value of N, P, f_S, K_{PD} and K_{VCO}, (that are defined for a specific application) we can calculate the desired loop parameters, ω_1 and ϕ_M, to obtain the loop filter coefficients of the digital loop filter. Once the loop filter coefficients are established, the open loop response is completely defined. We observe that β is a scaled *tangent* function, where ϕ (phase margin) is the argument to the tangent function and both P and ω_c serve as scale factors. So, for a given phase margin requirement, β scales linearly with the product of P and ω_c. Furthermore, both of the expressions for α and γ are functions of ϕ. We note that α and γ may be expressed as scaled versions of β. We can define a new function of ϕ:

$$f(\varphi) = 1 + \frac{1}{\sin(\varphi)} \qquad (20)$$

Then we can express γ as:

$$\gamma = \frac{f(\varphi)}{2} \beta \qquad (21)$$

In a similar manner, α can be expressed as:

$$\alpha = \frac{P \omega_c^2 N}{K_{PD} K_{VCO}} f(\varphi) \beta \qquad (22)$$

With the open loop response determined, the next step is to analyze the closed loop performance. The closed loop response is given by:

$$H_{CL} = \frac{G}{1+H_{OL}} = \frac{\left(K_{PD} H_{LF} \frac{K_{VCO}}{z-1} \right)}{1 + \left(K_{PD} H_{LF} \frac{K_{VCO}}{z-1} \right) (K_{DIV})} \qquad (23)$$

7. RESULTS

Proposed D-PLL is integrated in a digital device that that provides an output frequency from 8 kHz to 400 MHz. The output signal – that is synchronized to an input reference - is used for synchronization for various systems, including synchronous optical networks (SONET/SDH) and wireless infrastructure transceivers (GSM/UMTS) [15]. Figure 7

shows the digital PLL closed loop gain response (y-axis) versus frequency (x-axis). The D-PLL reference input is 50 MHz sinusoidal signal, the system clock is 800 MHz, the DDS output frequency is 200 MHz and the phase frequency detector operates at 1 MHz. The loop filter bandwidth is set to 6 kHz and the phase margin to 60°. The blue line is the measured gain response in the laboratory. The red line is D-PLL fixed point simulated gain response and the green line is gain response of the equivalent analog PLL. Simple observation shows that lab measurements match very well with simulation.

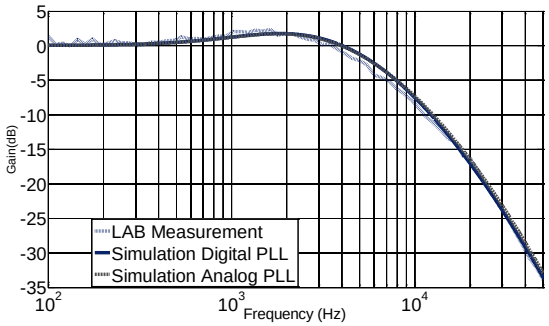


Figure 7. PLL close loop gain response.

8. CONCLUSIONS

A DDS based digital phase locked loop is presented. Proposed loop filter has a digital filter equivalent to a second order low-pass analog filter and a second order Cascade Comb-Integrator re-sampling filter that allows a loop bandwidth configuration from 1 mHz to 100 kHz. Loop filter analysis is presented and simulated closed loop gain response is compared to measured closed loop gain response on a laboratory bench.

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